AI-Driven Multi-Objective Optimization for Sustainable Construction Cost and Carbon Reduction

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Abstract

The construction industry significantly contributes to global carbon emissions and resource consumption. With increasing environmental regulations and sustainability targets, there is a critical need for intelligent systems that optimize both cost and environmental impact. This paper proposes an AI-driven multi-objective optimization framework applied to Bill of Quantities (BOQ) datasets for sustainable construction planning. The framework integrates emission factor mapping, carbon computation, and cost modeling with evolutionary optimization techniques such as NSGA-II to generate Pareto-optimal trade-offs between economic and environmental objectives. The study demonstrates how structured BOQ datasets can be transformed into actionable decision-making tools. Experimental results show that the proposed system effectively identifies material substitutions and quantity adjustments that reduce embodied carbon while maintaining acceptable cost limits. The approach highlights the potential of artificial intelligence in advancing sustainable engineering practices.

Keywords: Artificial Intelligence, Multi-Objective Optimization, Sustainability, Carbon Emissions, Construction Management, NSGA-II, BOQ Analysis

1 Introduction

The construction sector accounts for a substantial percentage of global greenhouse gas emissions, largely due to embodied carbon in materials such as cement, steel, and bricks. Traditional construction planning primarily focuses on cost efficiency and structural safety, often overlooking environmental implications. With growing climate concerns and regulatory frameworks targeting net-zero emissions, integrating sustainability into early-stage project planning has become essential.

Bill of Quantities (BOQ) documents provide detailed listings of materials, quantities, and costs for construction projects. However, these documents are rarely used as analytical datasets for environmental optimization. Transforming

BOQ data into structured datasets enables the application of machine learning and optimization algorithms to make informed decisions.

This study proposes an AI-based framework that converts raw BOQ data into a structured dataset, maps materials to standardized emission factors, computes embodied carbon per item, and performs multi-objective optimization to balance cost and carbon reduction. The goal is to develop a scalable, research-level approach that aligns economic feasibility with sustainability objectives.

2 Methodology

Method details.

3 Results

Results here.

4 Conclusion

Final thoughts.

References

[1] Sample Ref.